DAT151: Assignment 3 Report

Group: 5

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Task 1: SELinux

Objective: Work with SELinux contexts, users, domains, and policies.

1a: SELinux Status

Task: Check whether SELinux is currently enabled and make sure it is in targeted and enforcing mode.

- **Commands Used:**
- `sestatus`
- **Terminal Output / Printout:**

```
ffeilke@host22:~$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
ffeilke@host22:~$
```

- **Explanation:** Under SELinux status we see that SELinux is already enabled and under Current mode we check that we are in enforcing Mode.The loaded policy name is also set to targeted. In action SELinux consults its Targeted Policy database. It asks: "Is a process with label A allowed to perform action X on a file with label B. The enforcing mode ensures that unallowed operations will be blocked.

1b: Working with SELinux Users

- **1b.1 Mapping Linux users to SELinux users:**

Task: List the mapping between Linux users and SELinux confined users on your computer.

- **Commands used:**
- `sudo semanage login -l`

- **Output:**

```
ffeilke@host22:~$ sudo semanage login -l
[sudo] password for ffeilke:

Login Name           SELinux User         MLS/MCS Range        Service

__default__          unconfined_u         s0-s0:c0.c1023       *
restricteduser       user_u               s0                   *
root                 unconfined_u         s0-s0:c0.c1023       *
ffeilke@host22:~$
```

- **Explanation:**

- Any new user you create will automatically be mapped to the SELinux user listed under default (unconfined_u). SELinux doesn't interrupt daily tasks but still protects specific network services.

- **1b.2 Available SELinux Users:**

Task: List all available SELinux users on your computer using `seinfo`.

- **Commands used:**

- `seinfo -u`

- **Output:**

```
ffeilke@host22:~$ seinfo -u

Users: 8
   guest_u
   root
   staff_u
   sysadm_u
   system_u
   unconfined_u
   user_u
   xguest_u
ffeilke@host22:~$
```

- **1b.3 Restricting `su` and `sudo`:**

Task: Create a new Linux user and use SELinux to prevent this user from using the `su` and `sudo` tools.

- **Commands & Output:**

```
ffeilke@host22:~$ sudo useradd restricteduser2
[sudo] password for ffeilke:
Sorry, try again.
[sudo] password for ffeilke:
Sorry, try again.
[sudo] password for ffeilke:
ffeilke@host22:~$ sudo passwd restricteduser2
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
```

```
ffeilke@host22:~$ sudo semanage login -a -s user_u restricteduser2
ffeilke@host22:~$ sudo semanage login -l | grep restricteduser2
restricteduser2      user_u               s0                   *
ffeilke@host22:~$
```

- **Explanation of Method:**
- We change the restricted User from a unconfined_u to a user_u
- _t (Type Enforcement)s the primary way permissions are handled
- The tools su and sudo are setuid binaries -> when a user runs them, the process attempts to change its Effective User ID to root (UID 0) and SELinux blocks this for user_u

1c.

Domain Transitions (systemd to Apache)

Task: Verify that an Apache web server started by systemd has read access to the content of `/var/www/html`. Note: For each step below, you must use the `sesearch` command and document the applicable policy rules.

1c.1 Is systemd allowed to start the Apache web server?

- SELinux type of systemd: init_t
- SELinux type of Apache executable: httpd_exec_t
- Is systemd allowed to run it?: Yes

Commands & Policy Output (`sesearch`):

```
ffeilke@host22:~$ sesearch -s init_t -t httpd_exec_t -c file -p execute --allow
allow initrc_domain direct_init_entry:file { execute getattr map open read };
ffeilke@host22:~$ ps -p 1 -Z
LABEL                             PID TTY          TIME CMD
system_u:system_r:init_t:s0         1 ?        00:00:02 systemd
ffeilke@host22:~$ ls -Z /usr/sbin/httpd
system_u:object_r:httpd_exec_t:s0 /usr/sbin/httpd
ffeilke@host22:~$
```

- This rule explicitly allows the init_t domain to perform the execute operation on files labeled httpd_exec_t

1c.2 Can the Apache web server run in domain `httpd_t`?

Commands & Policy Output (`sesearch`):

```
ffeilke@host22:~$ sesearch -s init_t -t httpd_exec_t --type_transition
type_transition init_t httpd_exec_t:process httpd_t;
ffeilke@host22:~$ sesearch -s httpd_t -t httpd_exec_t -c file -p entrypoint --allow
allow httpd_t httpd_exec_t:file { entrypoint execute execute_no_trans getattr ioctl lock map open r
```

- When a process in init_t executes a file of type httpd_exec_t, the resulting process should run in the httpd_t domain

1c.3 Is systemd allowed a transition to `httpd_t`?

Commands & Policy Output (`sesearch`):

```
ffeilke@host22:~$ sesearch -s init_t -t httpd_t -c process -p transition --allow
allow initrc_domain daemon:process transition;
```

- This rule grants init_t the permission to actually transition the process context into httpd_t
-

1c.4 Has domain `httpd_t` access to open and read files in `/var/www/html`?

Commands & Policy Output (`sesearch`):

```
ffeilke@host22:~$ sesearch -s httpd_t -t httpd_sys_content_t -c file -p read --allow
allow httpd_t httpd_content_type:file { getattr ioctl lock map open read };
allow httpd_t httpdcontent:file { append create getattr ioctl link lock open read rename setattr unlink watch watch_reads write }; [ ( httpd_builtin_scripting && httpd_unified && httpd_enable_cgi ) ]:True
allow httpd_t httpdcontent:file { execute getattr ioctl map open read }; [ ( httpd_builtin_scripting && httpd_unified && httpd_enable_cgi ) ]:True
```

- This rule allows the Apache domain (httpd_t) to open and read files labeled as system web content (httpd_sys_content_t)

1d.

SELinux Boolean for `public_html`

Task: Use a SELinux boolean to allow Apache to read web content in a `public_html` directory in the home directory of users.

- **Commands & Output:**

```
ffeilke@host22:~$ man httpd_selinux
ffeilke@host22:~$ sudo setsebool -P httpd_enable_homedirs on
[sudo] password for ffeilke:
ffeilke@host22:~$ getsbool httpd_enable_homedirs
bash: getsbool: command not found...
ffeilke@host22:~$ getsebool httpd_enable_homedirs
httpd_enable_homedirs --> on
```

1e.

Custom Web Directory `/www`

Task: Create a directory `/www` and configure SELinux to allow Apache to read web content in this directory.

- **Commands & Output:**

```
ffeilke@host22:~$ sudo mkdir /www
echo "test /www" | sudo tee /www/index.html
[sudo] password for ffeilke:
test /www
ffeilke@host22:~$ sudo semanage fcontext -a -t httpd_sys_content_t "/www(/.*)?"
ffeilke@host22:~$ sudo restorecon -Rv /www
Relabeled /www from unconfined_u:object_r:default_t:s0 to unconfined_u:object_r:httpd_sys_content_t:s0
Relabeled /www/index.html from unconfined_u:object_r:default_t:s0 to unconfined_u:object_r:httpd_sys_content_t:s0
ffeilke@host22:~$ sesearch -s httpd_t -t httpd_sys_content_t -c file -p read --allow
allow httpd_t httpd_content_type:file { getattr ioctl lock map open read };
allow httpd_t httpdcontent:file { append create getattr ioctl link lock open read rename setattr unlink watch watch_r
eads write }; [ ( httpd_builtin_scripting && httpd_unified && httpd_enable_cgi ) ]:True
allow httpd_t httpdcontent:file { execute getattr ioctl map open read }; [ ( httpd_builtin_scripting && httpd_unified
 && httpd_enable_cgi ) ]:True
```

Task 2: Printing

Task: Install CUPS and all necessary dependencies, start the cups service, set it to run on startup, and add the HVL printing system.

- **Commands & Output (Installation & Setup):**

```
ffeilke@host22:~$ sudo dnf install cups cups-filters -y
Last metadata expiration check: 20:13:03 ago on Thu 12 Mar 2026 05:10:08 PM CET.
Package cups-1:2.4.10-12.el10_1.2.x86_64 is already installed.
Package cups-filters-1:2.0.0-10.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
ffeilke@host22:~$ sudo systemctl enable --now cups
ffeilke@host22:~$ systemctl status cups
● cups.service - CUPS Scheduler
     Loaded: loaded (/usr/lib/systemd/system/cups.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-03-13 13:02:38 CET; 21min ago
 Invocation: 10b976d421364ee0ad449b099533775e
TriggeredBy: ● cups.path
             ● cups.socket
       Docs: man:cupsd(8)
   Main PID: 1459 (cupsd)
     Status: "Scheduler is running..."
      Tasks: 1 (limit: 46636)
     Memory: 4.7M (peak: 4.9M)
        CPU: 28ms
     CGroup: /system.slice/cups.service
             └─1459 /usr/sbin/cupsd -l
```

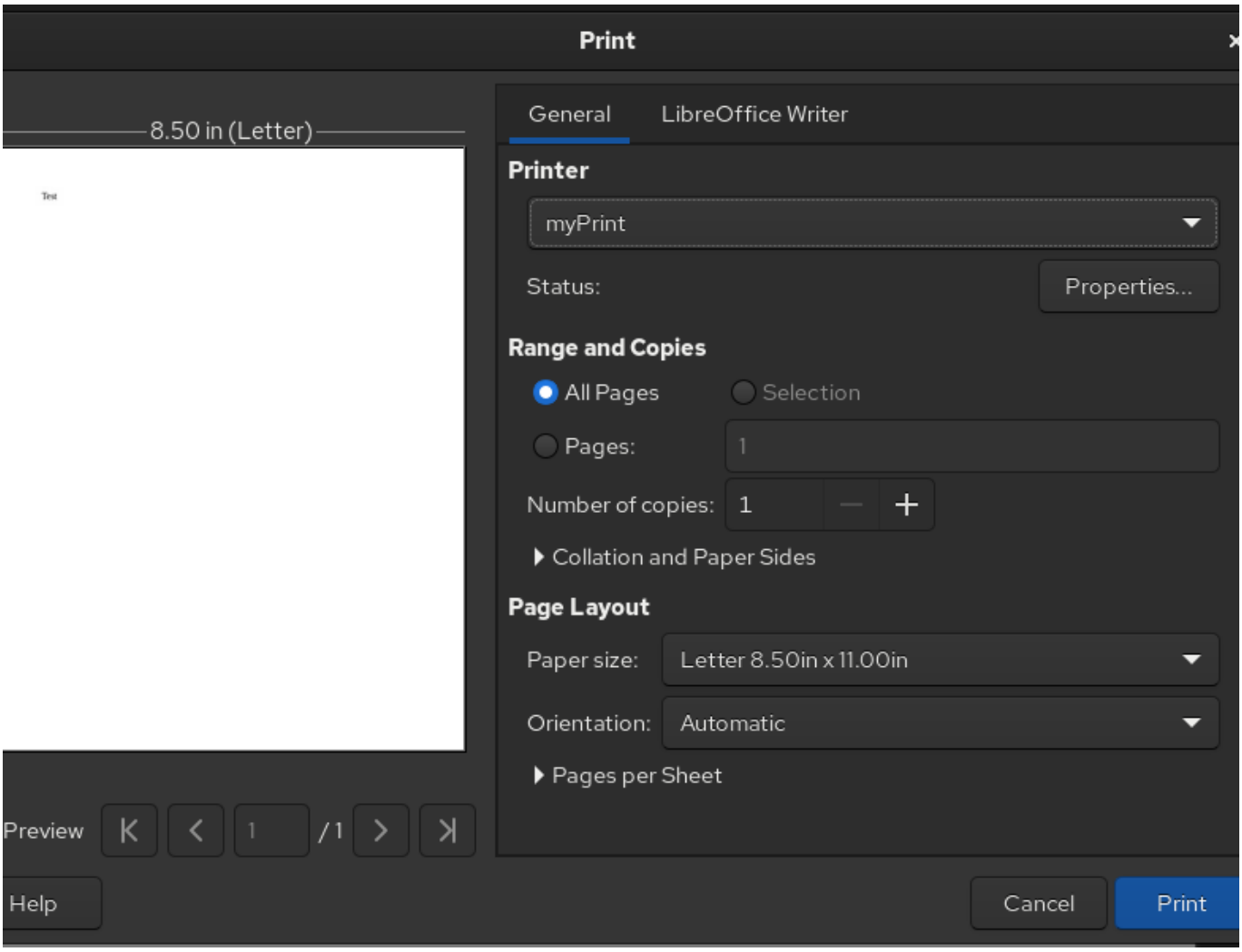
- cd '/home/ffeilke/Downloads/myPrintInstaller_LINUX'
- './installprinter.sh'

```
--Printer Installer--
Installing printer...Please wait...
[sudo] password for ffeilke:
lpadmin: Printer drivers are deprecated and will stop working in a future version of CUPS.

--Finished installing printer--
```

2.1 Print job from LibreOffice:

- cd /home/ffeilke/Downloads/LibreOffice_26.2.1_Linux_x86-64_rpm/LibreOffice_26.2.1.2_Linux_x86-64_rpm/RPMS
- sudo dnf install *.rpm -y

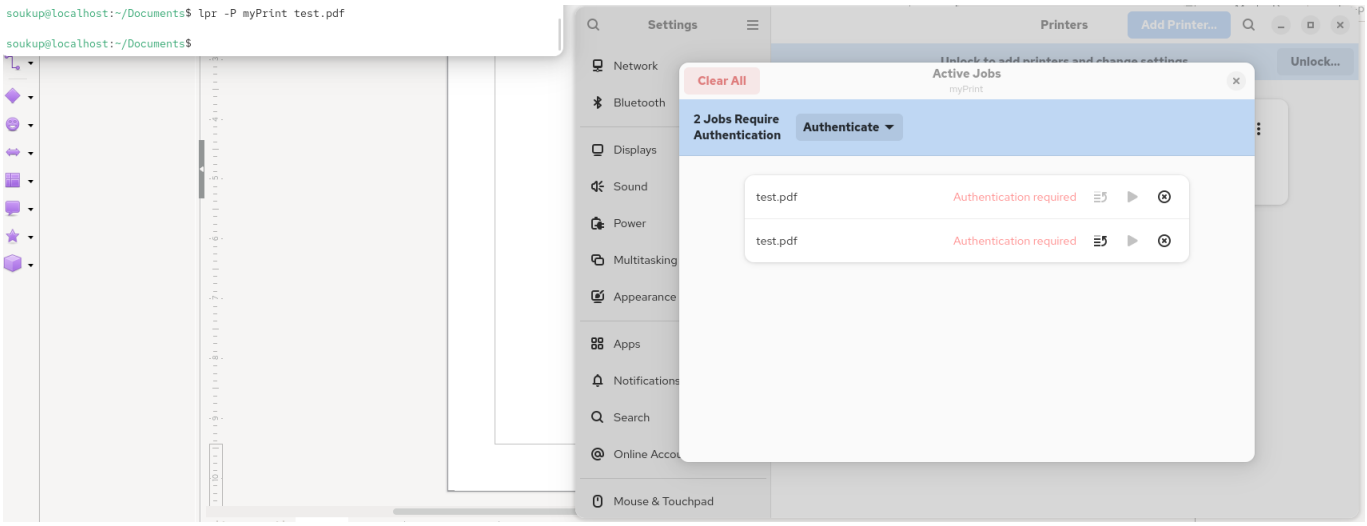


2.2 Print job from command line (`lpr`):

- **Commands & Output:**

```
ffeilke@host22:~$ lpr -P myPrint /home/ffeilke/Documents/testprint.odt
ffeilke@host22:~$ lpq -P myPrint
myPrint is ready
no entries
```

Fabienne didn't have any issues, I had a small one with authentication setup as I didn't input the credentials at the right time, but in the end with the help of the lecturer we made it work as well. We tried multiple commands to change the credentials, which can be seen in the following screenshot.




```
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v "https://jasou2367:I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g@myprin
t.hvl.no/RicohmyPrint/printers/ipp/print"
bash: !q: event not found
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://jasou2367:'I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g'@mypri
nt.hvl.no/RicohmyPrint/printers/ipp/print
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/printers/ip
p/print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://'jasou2367':'I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g'@myp
rint.hvl.no/RicohmyPrint/printers/ipp/print
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/printers/ip
p/print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://jasou2367:I2qVAl3@Y9Xi\!q$5dkJgXZ%PNc1c*g@myprin
t.hvl.no/RicohmyPrint/printers/ipp/print
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3@Y9Xi!qdkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/printers/ipp/
print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://jasou2367:I2qVAl3\@Y9Xi\!q\$5dkJgXZ\%PNc1c\*g@my
print.hvl.no/RicohmyPrint/printers/ipp/print
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/printers/ip
p/print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://jasou2367\:I2qVAl3\@Y9Xi\!q\$5dkJgXZ\%PNc1c\*g@m
yprint.hvl.no/RicohmyPrint/printers/ipp/print
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/printers/ip
p/print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://jasou2367:I2qVAl3\@Y9Xi\!q\$5dkJgXZ\%PNc1c\*g\@m
yprint.hvl.no/RicohmyPrint/printers/ipp/print
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/printers/ip
p/print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://jasou2367:I2qVAl3%40Y9Xi%21q%245dkJgXZ%PNc1c*g@m
yprint.hvl.no/RicohmyPrint/printers/ipp/print
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3%40Y9Xi%21q%245dkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/print
ers/ipp/print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v "https://jasou2367:I2qVAl3%40Y9Xi%21q%245dkJgXZ%PNc1c*g@
myprint.hvl.no/RicohmyPrint/printers/ipp/print"
lpadmin: Bad device-uri "https://jasou2367:I2qVAl3%40Y9Xi%21q%245dkJgXZ%PNc1c*g@myprint.hvl.no/RicohmyPrint/print
ers/ipp/print".
soukup@localhost:~/Downloads$ sudo lpadmin -p myPrint -v https://'jasou2367':'I2qVAl3@Y9Xi!q$5dkJgXZ%PNc1c*g'@myp
rint.hvl.no/RicohmyPrint/printers/ipp/print
```

Task 3: Open Ports and Processes

Task: Find all ports open for tcp and udp connections on your computer, and list the processes and services that use these ports.

- sudo ss -tulpn
- -t: Display TCP sockets

- -u: Display UDP sockets
- -l: Show only listening sockets (ports waiting for a connection)
- -p: Show the process name and PID using the socket
- -n: Show numerical port numbers

```
ffeilke@host22:~$ sudo ss -tulpn
[sudo] password for ffeilke:
Netid     State      Recv-Q     Send-Q                          Local Address:Port           Peer Address:Port
Process
udp       UNCONN     0          0                                     0.0.0.0:5353                0.0.0.0:*
 users:(("avahi-daemon",pid=1241,fd=12))
udp       UNCONN     0          0                                     0.0.0.0:39389               0.0.0.0:*
 users:(("wsdd",pid=8781,fd=8))
udp       UNCONN     0          0                                    10.0.0.67:3702               0.0.0.0:*
 users:(("wsdd",pid=8781,fd=9))
udp       UNCONN     0          0                              239.255.255.250:3702               0.0.0.0:*
 users:(("wsdd",pid=8781,fd=7))
udp       UNCONN     0          0                                    127.0.0.1:323                0.0.0.0:*
 users:(("chronyd",pid=1293,fd=5))
udp       UNCONN     0          0                                        [::]:5353                   [::]:*
 users:(("avahi-daemon",pid=1241,fd=13))
udp       UNCONN     0          0                                           *:34194                     *:*
 users:(("wsdd",pid=8781,fd=11))
udp       UNCONN     0          0              [fe80::42a8:f0ff:fea3:dd8]%eno1:3702                   [::]:*
 users:(("wsdd",pid=8781,fd=12))
udp       UNCONN     0          0                              [ff02::c]%eno1:3702                   [::]:*
 users:(("wsdd",pid=8781,fd=10))
udp       UNCONN     0          0                                       [::1]:323                    [::]:*
 users:(("chronyd",pid=1293,fd=6))
tcp       LISTEN     0          4096                                 127.0.0.1:631                0.0.0.0:*
 users:(("cupsd",pid=1459,fd=9))
tcp       LISTEN     0          80                                    0.0.0.0:3306                0.0.0.0:*
 users:(("mariadbd",pid=1551,fd=21))
tcp       LISTEN     0          128                                   0.0.0.0:22                  0.0.0.0:*
 users:(("sshd",pid=1468,fd=7))
tcp       LISTEN     0          80                                       [::]:3306                   [::]:*
 users:(("mariadbd",pid=1551,fd=22))
tcp       LISTEN     0          4096                                        *:9090                     *:*
 users:(("systemd",pid=1,fd=464))
tcp       LISTEN     0          4096                                    [::1]:631                    [::]:*
 users:(("cupsd",pid=1459,fd=8))
tcp       LISTEN     0          128                                      [::]:22                     [::]:*
 users:(("sshd",pid=1468,fd=8))
```

- **Explanation of Findings:**
- CUPS - Port 631 (TCP)
- Avahi - Port 5353(UDP)
- WSDD - Ports 3702, 39389, 34194 (UDP)
- SSH - Port 22 (TCP)
- Cockpit - Port 9090 (TCP)
- MariaDB - Port 3306 (TCP)
- Coronyd -Port 323 (UDP)
- Netid shows the type of protocoll

- State shows the status of that port
- Recv-Q shows the amount of data (in bytes) received but not yet processed by the program
- Send-Q shows the amount of data sent that hasn't been acknowledged by the other side yet
- Local Address:Port defines the internal IP address and unique port number on the local machine where a specific service is hosted and listening for traffic
- Peer Address:Port dentifies the remote IP address and port number of the external device that is either currently communicating with the local service or is permitted to connect to it

Task 4: Two-Factor Authentication (2FA)

> **Important:** Do not modify any of the files below `/etc/ssh/`; create new files with the modifications.

- **Task:** Set up 2FA for SSH login on your computer, using an SSH key and time-based one-time passwords. Set this up on all lab computers of the group.

```
soukup@localhost:~$ sudo dnf install google-authenticator -y
Last metadata expiration check: 1:13:53 ago on Mon 02 Mar 2026 12:26:35 PM CET.
Dependencies resolved.
================================================================================================================
 Package                              Architecture          Version                       Repository          Size
================================================================================================================
Installing:
 google-authenticator                 x86_64                1.11-1.el10_1                 epel                 56 k

Transaction Summary
================================================================================================================
Install  1 Package

Total download size: 56 k
Installed size: 118 k
Downloading Packages:
google-authenticator-1.11-1.el10_1.x86_64.rpm                                      125 kB/s |  56 kB     00:00
----------------------------------------------------------------------------------------------------------------
Total                                                                               46 kB/s |  56 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                        1/1
  Installing       : google-authenticator-1.11-1.el10_1.x86_64                                              1/1
  Running scriptlet: google-authenticator-1.11-1.el10_1.x86_64                                              1/1

Installed:
  google-authenticator-1.11-1.el10_1.x86_64

Complete!
soukup@localhost:~$ touch /etc/ssh/sshd^C
soukup@localhost:~$ google-authenticator -t -d -f -r 3 -R 30 -W -s ~/.ssh/.google_authenticator
Failed to use libqrencode to show QR code visually for scanning.
Consider typing the OTP secret into your app manually.
Your new secret key is: E433A73PJKZWM6X4MKARJOIJEXGKOECX
Enter code from app (-1 to skip): ^C
soukup@localhost:~$ sudo dnf install qrencode
Last metadata expiration check: 1:17:50 ago on Mon 02 Mar 2026 12:26:35 PM CET.
Dependencies resolved.
================================================================================================================
 Package                          Architecture          Version                       Repository          Size
================================================================================================================
Installing:
 qrencode                         x86_64                4.1.1-9.el10_0                epel                 25 k
```

First we installed the google authenticator and qrencode packages

```
soukup@localhost:~$ mkdir -p ~/.ssh
soukup@localhost:~$ ls .ssh/
soukup@localhost:~$ chmod 700 ~/.ssh
soukup@localhost:~$ google-authenticator -t -d -f -r 3 -R 30 -W -s ~/.ssh/.google_authenticator
```

```
Your new secret key is: PIUDC5XT346MUWSL353Y3GWNDX3I5WJU
Enter code from app (-1 to skip): 264724
Code confirmed
Your emergency scratch codes are:
  18866582
  46236224
  64181750
  54771316
  47169358
soukup@localhost:~$
```

Command to generate the secret and `google-authenticator` showing the generated secret and QR code for enrollment.

```
soukup@localhost:/etc/ssh$ cd ssh_config.d/
soukup@localhost:/etc/ssh/ssh_config.d$ ls -la
total 8
drwxr-xr-x. 2 root root   28 Jan 19 12:42 .
drwxr-xr-x. 4 root root 4096 Jan 19 11:48 ..
-rw-r--r--. 1 root root  581 Dec 17 01:00 50-redhat.conf
soukup@localhost:/etc/ssh/ssh_config.d$ sudo nano 99-2fa.conf
soukup@localhost:/etc/ssh/ssh_config.d$ sudo nano 99-2fa.conf
soukup@localhost:/etc/ssh/ssh_config.d$ sudo systemctl restart sshd
soukup@localhost:/etc/ssh/ssh_config.d$
```

```
soukup@localhost:/etc/ssh/ssh_config.d – sudo nano 99-2fa.conf
/etc/ssh/ssh_config.d
  GNU nano 8.1                                99-2fa.conf
KbdInteractiveAuthentication yes
PasswordAuthentication no
AuthenticationMethods publickey,keyboard-interactive
```

- Then, we created a new file `/etc/ssh/sshd_config.d/99-2fa.conf` to setup ssh config properly for the google authenticator
  - KbdInteractiveAuthentication - allows the server to prompt for the time-based one-time password (TOTP) during the login process
  - PasswordAuthentication - disable simple password login
  - AuthenticationMethods - specify what methods are used to login

```
soukup@localhost:~$ sudo nano /etc/pam.d/sshd
soukup@localhost:~$ sudo systemctl restart sshd
soukup@localhost:~$
```

```
  GNU nano 8.1                                /etc/pam.d/sshd
#%PAM-1.0
auth       substack     password-auth
#google oauth setup
auth required pam_google_authenticator.so secret=/home/${USER}/.ssh/.google_authenticator
auth       include      postlogin
account    required     pam_sepermit.so
account    required     pam_nologin.so
account    include      password-auth
password   include      password-auth
# pam_selinux.so close should be the first session rule
session    required     pam_selinux.so close
session    required     pam_loginuid.so
# pam_selinux.so open should only be followed by sessions to be executed in the user context
session    required     pam_selinux.so open env_params
session    required     pam_namespace.so
session    optional     pam_keyinit.so force revoke
session    optional     pam_motd.so
session    include      password-auth
session    include      postlogin
```

Then we configured the /etc/pam.d/sshd config file to use the google authenticator

```
soukup@localhost:~$ nano ~/.ssh/authorized_keys
soukup@localhost:~$ chmod 600 ~/.ssh/authorized_keys
soukup@localhost:~$ restorecon -Rv ~/.ssh/
```

```
john@localhost:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/john/.ssh/id_ed25519):
Enter passphrase for "/home/john/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/john/.ssh/id_ed25519
Your public key has been saved in /home/john/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:2jFjqsr8o7uZcL/gmV/zF2e/JsttYWEyPmrbAQo94AE john@localhost.localdomain
The key's randomart image is:
+--[ED25519 256]--+
|    E            |
|     .           |
|      o          |
|     . +    o o  |
|      o S .. + . |
|       * *..= o  |
|. o   = o  =.+ . |
| * B.o o  +o.o+  |
|  &B*o  .o..+=o. |
+----[SHA256]-----+
john@localhost:~$ ls -la
total 20
drwx------. 4 john john 125 Mar  2 14:12 .
drwxr-xr-x. 4 root root  32 Mar  2 12:15 ..
-rw-------. 1 john john  97 Mar  2 14:12 .bash_history
-rw-r--r--. 1 john john  18 Oct 29  2024 .bash_logout
-rw-r--r--. 1 john john 144 Oct 29  2024 .bash_profile
-rw-r--r--. 1 john john 522 Oct 29  2024 .bashrc
-rw-r--r--. 1 john john 334 Jun 17  2025 .emacs
drwxr-xr-x. 4 john john  39 Jan 19 12:42 .mozilla
drwx------. 2 john john  88 Mar  9 12:06 .ssh
john@localhost:~$ cat .ssh/
id_ed25519       id_ed25519.pub   known_hosts      known_hosts.old
john@localhost:~$ cat .ssh/
id_ed25519       id_ed25519.pub   known_hosts      known_hosts.old
john@localhost:~$ cat .ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJPqigycNQ+P3akOxKDlE8x8KEKpZW94O7aXhfx2OzeO john@localhost.localdomain
john@localhost:~$ ssh soukup@10.0.0.68
(soukup@10.0.0.68) Password:
(soukup@10.0.0.68) Verification code:

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

Web console: https://localhost:9090/ or https://10.0.0.68:9090/

Last login: Mon Mar  9 12:02:18 2026
soukup@localhost:~$
```

We created new key pair fot the local users, added them to `.ssh` to be available. We were then able to login to the other user account with the google authentication - using the code in the Authenticator app and password as a secure login.

Task 5: Secure your computer

Task: Use Fail2Ban to secure your computer. Set PAM requirements for password qualities, password history, and account locking if there are many consecutive authentication failures.

> **Requirement:** Do not modify directly the PAM files; configure the PAM system using `authselect`.

- **Commands & Output:**

We installed and enabled Fail2Ban, and used `authselect` to enable PAM features such as account lockout and password history. Below are all Task 5 screenshots with short captions. Each image is followed by one blank line and then its caption.

Screenshots:

```
333  sudo dnf install fail2ban
334  sudo systemctl enable --now fail2ban
```

Fail2Ban installed

```
soukup@localhost:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-03-09 12:13:04 CET; 29s ago
 Invocation: f69e4bfb71ac43b485df9a3b39eaa056
       Docs: man:fail2ban(1)
    Process: 8348 ExecStartPre=/bin/mkdir -p /run/fail2ban (code=exited, status=0/SUCCESS)
   Main PID: 8350 (fail2ban-server)
      Tasks: 3 (limit: 46652)
     Memory: 11.5M (peak: 12.2M)
        CPU: 93ms
     CGroup: /system.slice/fail2ban.service
             └─8350 /usr/bin/python3 -sP /usr/bin/fail2ban-server -xf start

Mar 09 12:13:04 localhost.localdomain systemd[1]: Starting fail2ban.service - Fail2Ban Service...
Mar 09 12:13:04 localhost.localdomain systemd[1]: Started fail2ban.service - Fail2Ban Service.
Mar 09 12:13:04 localhost.localdomain fail2ban-server[8350]: Server ready
```

Running fail2ban service

```
soukup@localhost:~$ authselect current
Profile ID: local
Enabled features:
- with-fingerprint
soukup@localhost:~$ sudo authselect list
- local          Local users only
- sssd           Enable SSSD for system authentication (also for local users only)
- winbind        Enable winbind for system authentication
soukup@localhost:~$ sudo authselect select sssd with-faillock with-passwd-history
[error] Unknown profile feature [with-passwd-history], did you mean [with-pwhistory]?
[error] Unable to activate profile [sssd] [22]: Invalid argument
Unable to activate profile [22]: Invalid argument
soukup@localhost:~$ sudo authselect select sssd with-faillock with-pwhistory
Profile "sssd" was selected.

Make sure that SSSD service is configured and enabled. See SSSD documentation for more information

soukup@localhost:~$ sudo authselect apply-changes
Changes were successfully applied.
soukup@localhost:~$ sudo authselect current
Profile ID: sssd
Enabled features:
- with-faillock
- with-pwhistory
```

`authselect` profile selection and feature enabling (faillock, passwd-history).

```
soukup@localhost:~ — sudo nano /etc/security/faillock.conf
  GNU nano 8.1                                                    /etc/security/faillock.conf
# Configuration for locking the user after multiple failed
# authentication attempts.
#
# The directory where the user files with the failure records are kept.
# The default is /var/run/faillock.
# dir = /var/run/faillock
#
# Will log the user name into the system log if the user is not found.
# Enabled if option is present.
# audit
#
# Don't print informative messages.
# Enabled if option is present.
# silent
#
# Don't log informative messages via syslog.
# Enabled if option is present.
# no_log_info
#
# Only track failed user authentications attempts for local users
# in /etc/passwd and ignore centralized (AD, IdM, LDAP, etc.) users.
# The `faillock` command will also no longer track user failed
# authentication attempts. Enabling this option will prevent a
# double-lockout scenario where a user is locked out locally and
# in the centralized mechanism.
# Enabled if option is present.
# local_users_only
#
# Deny access if the number of consecutive authentication failures
# for this user during the recent interval exceeds n tries.
# The default is 3.
deny = 3
#
# The length of the interval during which the consecutive
# authentication failures must happen for the user account
# lock out is <replaceable>n</replaceable> seconds.
# The default is 900 (15 minutes).
fail_interval = 90
#
# The access will be re-enabled after n seconds after the lock out.
# The value 0 has the same meaning as value `never` - the access
# will not be re-enabled without resetting the faillock
# entries by the `faillock` command.
# The default is 600 (10 minutes).
unlock_time = 60
#
# Root account can become locked as well as regular accounts.
# Enabled if option is present.
```

Edited faillock options to be more user friendly with wrong password inputs.

```
soukup@localhost:~ — sudo nano /etc/security/pwhistory.conf
  GNU nano 8.1                                                    /etc/security/pwhistory.conf
# Configuration for remembering the last passwords used by a user.
#
# Enable the debugging logs.
# Enabled if option is present.
# debug
#
# root account's passwords are also remembered.
# Enabled if option is present.
# enforce_for_root
#
# Number of passwords to remember.
# The default is 10.
remember = 5
#
# Number of times to prompt for the password.
# The default is 1.
retry = 1
#
# The file where the last passwords are kept.
# The default is /etc/security/opasswd.
# file = /etc/security/opasswd
```

Password history file edited.

```
soukup@localhost:~ — sudo nano /etc/security/pwquality.conf

  GNU nano 8.1                                                    /etc/security/pwquality.conf
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
# difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
minlen = 10
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
# dcredit = 0
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
# ucredit = 0
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
# lcredit = 0
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
# ocredit = 0
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
minclass = 3
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
maxrepeat = 3
#
# The maximum number of allowed consecutive characters of the same class in the
# new password.
# The check is disabled if the value is 0.
# maxclassrepeat = 0
#
# Whether to check for the words from the passwd entry GECOS string of the user.
# The check is enabled if the value is not 0.
# gecoscheck = 0
#
```

Password quality file edited.

```
john@localhost:~$ passwd
Current password:
New password:
BAD PASSWORD: The password is shorter than 10 characters
passwd: Authentication token manipulation error
passwd: password unchanged
john@localhost:~$ passwd
Current password:
New password:
BAD PASSWORD: The password contains less than 3 character classes
passwd: Authentication token manipulation error
passwd: password unchanged
john@localhost:~$
```

Test of password change showing enforcement of password policies. Tried a short 6 char password and then long password but withou enough classes, both got rejected.

Task 6: SSH

Setup: Create an SSH key pair as a normal user on your own computer and start an OpenSSH server on your lab computer.

6.1 Direct Login Attempt

Task: Try to log in to your lab computer from your own computer.

We attempted a direct SSH connection to the lab host but received a network-level failure (connection timed out). This indicates the lab host is not directly reachable from the client network (no SSH port open or firewall/gateway blocks direct access). The screenshots below show the failed connection and the local key setup.

```
❯ ssh soukup@10.0.0.68
ssh: connect to host 10.0.0.68 port 22: Connection timed out
~                                                    255 ✘ 〈 took 2m 13s 〈 at 12:29:00
```

Direct connection attempt showing network-level failure - connection timeout.

```
❯ ssh-keygen -t ed25519 -f ~/.ssh/DAT151_ed25519 -C "DAT151_Lab"
Generating public/private ed25519 key pair.
Enter passphrase for "/home/souki/.ssh/DAT151_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/souki/.ssh/DAT151_ed25519
Your public key has been saved in /home/souki/.ssh/DAT151_ed25519.pub
The key fingerprint is:
SHA256:xxFFZCda9UHFlMCuqGrLz/wj6YZk9yhfxINoc7vdUrw DAT151_Lab
The key's randomart image is:
+--[ED25519 256]--+
|          .+Oo===|
|           = + oo|
|          o .   .|
|       . o. . .  |
|      + oS+= .   |
|     .oo.oo.+    |
|     o o.=.. .   |
|     .+o*++.E    |
|     .+OBooo.    |
+----[SHA256]-----+
❯ eval "$(ssh-agent -s)"
ssh-add ~/.ssh/DAT151_ed25519
Agent pid 1510675
Identity added: /home/souki/.ssh/DAT151_ed25519 (DAT151_Lab)
```

SSH key generation and `ssh-copy-id` confirmation - this was later changed to use the provided `https://epleauth.azurewebsites.net` method, that already has public and secret keys generated and we just copied them.

6.2 Jump Host Login

Task: From your own computer, log into your lab computer with a jump through `eple.hvl.no`.

Using the gateway `eple.hvl.no` as a jump host allowed us to access because the gateway has network reachability to the lab host. We used `ssh -J` to route the connection through the gateway; the SSH key

authentication and agent forwarding were used to authenticate. Screenshots below show a successful jump-host login and an interactive shell on the lab host.

```
❯ ssh -J hvl@eple.hvl.no soukup@10.0.0.68
The authenticity of host 'eple.hvl.no (158.37.32.196)' can't be established.
ED25519 key fingerprint is SHA256:vYtMuEis9rWDlNsgA3vp36bA/vFUTk/4XZ+YKlAvY2I.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'eple.hvl.no' (ED25519) to the list of known hosts.
hvl@eple.hvl.no: Permission denied (publickey).
Connection closed by UNKNOWN port 65535

~                                                                    255 ✘ 〈 at 12:27:32
```

Tried connecting first, got rejected because of the key.

```
Connection closed by UNKNOWN port 65535
❯ ssh-keygen -t ed25519 -f ~/.ssh/DAT151_ed25519 -C "DAT151_Lab"
Generating public/private ed25519 key pair.
Enter passphrase for "/home/souki/.ssh/DAT151_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/souki/.ssh/DAT151_ed25519
Your public key has been saved in /home/souki/.ssh/DAT151_ed25519.pub
The key fingerprint is:
SHA256:xxFFZCda9UHFlMCuqGrLz/wj6YZk9yhfxINoc7vdUrw DAT151_Lab
The key's randomart image is:
+--[ED25519 256]--+
|          .+Oo===|
|           = + oo|
|          o .   .|
|       . o. . .  |
|      + oS+= .   |
|     .oo.oo.+    |
|    o o.=.. .    |
|    .+o*++.E     |
|    .+OBooo.     |
+----[SHA256]-----+
❯ eval "$(ssh-agent -s)"
ssh-add ~/.ssh/DAT151_ed25519
Agent pid 1510675
Identity added: /home/souki/.ssh/DAT151_ed25519 (DAT151_Lab)
❯ cat ~/.ssh/DAT151_ed25519
```

File: /home/souki/.ssh/DAT151_ed25519

```
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAAAMwAAAAtzc2gtZW
QyNTUxOQAAACCH/tgHlGpuzwE9GeFsh9/94UR3IFBBZ8BMRbpgrSfe3AAAAJDD0mlaw9Jp
WgAAAAtzc2gtZWQyNTUxOQAAACCH/tgHlGpuzwE9GeFsh9/94UR3IFBBZ8BMRbpgrSfe3A
AAAECRM1BqwBXsxDPUUUO3Ebi1Q+9s6dkzbb5CoaZx2cVk/Yf+2AeUam7PAT0Z4WyH3/3h
RHcgUEFnwExFumCtJ97cAAAACkRBVDE1MV9MYWIBAgM=
-----END OPENSSH PRIVATE KEY-----
```

```
❯ cat ~/.ssh/DAT151_ed25519.pub
```

File: /home/souki/.ssh/DAT151_ed25519.pub

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIf+2AeUam7PAT0Z4WyH3/3hRHcgUEFnwExFumCtJ97c DAT151_Lab
```

```
❯ ssh -J hvl@eple.hvl.no soukup@10.0.0.68
hvl@eple.hvl.no: Permission denied (publickey).
Connection closed by UNKNOWN port 65535
❯ ssh -J hvl@eple.hvl.no soukup@10.0.0.68
hvl@eple.hvl.no: Permission denied (publickey).
Connection closed by UNKNOWN port 65535
❯ ssh -i ~/.ssh/DAT151_ed25519 soukup@10.0.0.68
ssh: connect to host 10.0.0.68 port 22: Connection timed out
❯ ssh -J hvl@eple.hvl.no soukup@10.0.0.68
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@         WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0664 for '/home/souki/.ssh/id_rsa' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "/home/souki/.ssh/id_rsa": bad permissions
hvl@eple.hvl.no: Permission denied (publickey).
Connection closed by UNKNOWN port 65535
```

soukup@localhost:~ — nano /home/soukup/.ssh/authorized_keys

```
  GNU nano 8.1                    /home/soukup/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJPqigycNQ+P3akOxKDlE8x8KEKpZW9407aXhfx2OzeO john@localhost.localdomain
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIf+2AeUam7PAT0Z4WyH3/3hRHcgUEFnwExFumCtJ97c DAT151_Lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC07P7pQkSu/+mrXL9DDoEURmLCj+E0QozgE/bxom3KPSxhF7gEREzqDWsqv9EOCXomeQi1WlHw5Ig4
```

Tried viewing the key, adding it to ssh as possible solutioons, none worked. We also added the public key to the lab computer `authorized_keys` file. This didn't work, so we had to use the `eple.hvl.no` generated key.

```
❯ ls -la ~/.ssh/
total 68
drwx------  2 souki souki 4096 Mar  9 12:43 .
drwxr-x--- 49 souki souki 4096 Mar  9 12:43 ..
-rw-------  1 souki souki  399 Feb  6 12:17 ADA502-NREC
-rw-r--r--  1 souki souki   93 Feb  6 12:17 ADA502-NREC.pub
-rw-------  1 souki souki  399 Mar  9 12:29 DAT151_ed25519
-rw-r--r--  1 souki souki   92 Mar  9 12:29 DAT151_ed25519.pub
-rw-rw-r--  1 souki souki 1671 Feb  6 12:15 Soukup_Jan.pem
-rw-------  1 souki souki    0 Jun  6  2025 authorized_keys
-rw-------  1 souki souki  399 Jun  7  2025 cvut_ed25519
-rw-r--r--  1 souki souki   91 Jun  7  2025 cvut_ed25519.pub
-rw-------  1 souki souki  411 Jun  9  2025 github_personal_ed25519
-rw-r--r--  1 souki souki   97 Jun  9  2025 github_personal_ed25519.pub
-rw-rw-r--  1 souki souki  107 Feb  6 12:09 id_ed25519.pub
-rw-rw-r--  1 souki souki 1702 Mar  9 12:42 id_rsa
-rw-------  1 souki souki 3660 Mar  9 12:27 known_hosts
-rw-------  1 souki souki 2904 Feb  6 12:12 known_hosts.old
-rw-------  1 souki souki  399 Jun  9  2025 mcc_ed25519
-rw-r--r--  1 souki souki   88 Jun  9  2025 mcc_ed25519.pub
❯ chmod 600 ~/.ssh/id_rsa
❯ ls -la ~/.ssh/
total 68
drwx------  2 souki souki 4096 Mar  9 12:43 .
drwxr-x--- 49 souki souki 4096 Mar  9 12:44 ..
-rw-------  1 souki souki  399 Feb  6 12:17 ADA502-NREC
-rw-r--r--  1 souki souki   93 Feb  6 12:17 ADA502-NREC.pub
-rw-------  1 souki souki  399 Mar  9 12:29 DAT151_ed25519
-rw-r--r--  1 souki souki   92 Mar  9 12:29 DAT151_ed25519.pub
-rw-rw-r--  1 souki souki 1671 Feb  6 12:15 Soukup_Jan.pem
-rw-------  1 souki souki    0 Jun  6  2025 authorized_keys
-rw-------  1 souki souki  399 Jun  7  2025 cvut_ed25519
-rw-r--r--  1 souki souki   91 Jun  7  2025 cvut_ed25519.pub
-rw-------  1 souki souki  411 Jun  9  2025 github_personal_ed25519
-rw-r--r--  1 souki souki   97 Jun  9  2025 github_personal_ed25519.pub
-rw-rw-r--  1 souki souki  107 Feb  6 12:09 id_ed25519.pub
-rw-------  1 souki souki 1702 Mar  9 12:42 id_rsa
-rw-------  1 souki souki 3660 Mar  9 12:27 known_hosts
-rw-------  1 souki souki 2904 Feb  6 12:12 known_hosts.old
-rw-------  1 souki souki  399 Jun  9  2025 mcc_ed25519
-rw-r--r--  1 souki souki   88 Jun  9  2025 mcc_ed25519.pub
❯ ssh -J hvl@eple.hvl.no soukup@10.0.0.68
The authenticity of host '10.0.0.68 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:aWsnsc7J/XUBP8Wb1IPO9ZxGtGP7S0WFId6qXKMDmWA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.0.68' (ED25519) to the list of known hosts.
(soukup@10.0.0.68) Password:
(soukup@10.0.0.68) Verification code:
(soukup@10.0.0.68) Password:
(soukup@10.0.0.68) Verification code:

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

Activate the web console with: systemctl enable --now cockpit.socket

Last failed login: Mon Mar  9 12:44:56 CET 2026 from 10.0.0.36 on ssh:notty
There was 1 failed login attempt since the last successful login.
Last login: Mon Mar  9 12:08:47 2026 from 10.0.0.68
soukup@localhost:~$ rm ~/.ssh/
authorized_keys        .google_authenticator  known_hosts
```

Added the keys to private computer and lab computer as well. After that the connection was established.

6.3 MariaDB SSH Tunnel

Task: Install the MariaDB client tool on your own computer, set up an SSH tunnel through `eple.hvl.no` to the MariaDB server on your lab computer, and connect to it.

This task was more challenging then it should have been. We got stuck on the key authorization and discussed with the lecturer the solution for about an hour. In the end it was a combination of corrupted public key on the personal laptop and incorrect ssh calls to the lab computer. Following are screenshots showing the progress, issues we encountered, trying to find the solution and finally working MariaDB connection from personal computer. In the end we created a local port forward from `localhost:13389` to the remote MariaDB port on the lab host via the jump host and then connected the local MariaDB client to the forwarded port. The tunnel successfully forwarded traffic and the client connected to the remote database through the SSH tunnel.

```
❯ sudo apt install mariadb-client
[sudo: authenticate] Password:
The following packages were automatically installed and are no longer required:
  caca-utils  chafa  jp2a  libavif16  libchafa0t64  libgav1-1  libyuv0  nvidia-firmware-570-570.195.03  nvidia-firmware-580-580.95.05  toilet  toilet-fonts
Use 'sudo apt autoremove' to remove them.

Installing:
  mariadb-client

Installing dependencies:
  libconfig-inifiles-perl  libdbd-mariadb-perl  libdbi-perl  libmariadb3  libpcre2-posix3  libterm-readkey-perl  mariadb-client-core  mariadb-common  mysql-common
```

Installing the MariaDB client tool on personal computer.

```
❯ ssh -f -N -L 13389:10.0.0.68:3306 hvl@eple.hvl.no -vvv -A
debug1: OpenSSH_10.0p2 Ubuntu-5ubuntu5, OpenSSL 3.5.3 16 Sep 2025
debug3: Running on Linux 6.17.0-14-generic #14-Ubuntu SMP PREEMPT_DYNAMIC Fri Jan  9 17:01:16 UTC 2026 x86_64
debug3: Started with: ssh -f -N -L 13389:10.0.0.68:3306 hvl@eple.hvl.no -vvv -A
debug1: Reading configuration data /etc/ssh/ssh_config
```

Part of debug command we and lecturer used for finding out the solution to the connection.

```
❯ mariadb -h 127.0.0.1 -P 13389 -u soukup -p --ssl=0
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 18
Server version: 10.11.15-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> list users
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for th
e right syntax to use near 'list users' at line 1
MariaDB [(none)]> list databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for th
e right syntax to use near 'list databases' at line 1
MariaDB [(none)]> how databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for th
e right syntax to use near 'how databases' at line 1
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| PRIVBASE           |
| information_schema |
+--------------------+
2 rows in set (0.014 sec)

MariaDB [(none)]>
```

Finally working MariaDB client session over the tunnel after a lot of struggle and debugging.